

Motivation

Non-parametric testing

Wilcoxon rank-sum

Wilcoxon signed-rank

Kruskal-Wallis

Summary

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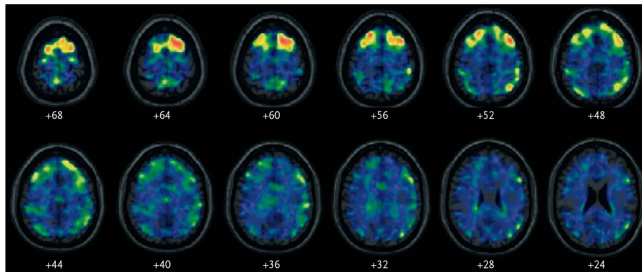
Kruskal-Wallis

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Abnormal Levels of a Protein Linked to C.T.E. Found in N.F.L. Players' Brains, Study Shows



NYtimes article

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ORIGINAL ARTICLE

Tau Positron-Emission Tomography in Former National Football League Players

Robert A. Stern, Ph.D., Charles H. Adler, M.D., Ph.D., Kewei Chen, Ph.D., Michael Navitsky, M.S., Ji Luo, M.S., David W. Dodick, M.D., Michael L. Allosco, Ph.D., Yorghos Tripodis, Ph.D., Dhruvan D. Goradia, Ph.D., Brett Martin, M.S., Diego Mastroeni, Ph.D., Nathan G. Fritts, B.A., [et al.](#)

Tau imaging is far from a diagnostic test for C.T.E. The results, published in *The New England Journal of Medicine*, are preliminary, but are a first step toward a clinical test for C.T.E. in living players.

From the article: statistical analysis

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Between-group comparisons of age, years of education, and MMSE scores were analyzed with **Mann-Whitney U tests**.

Group differences in race were analyzed with chi-square tests.

For amyloid-beta plaque burden, chi-square tests compared the proportion with a positive florbetapir PET, and **t-tests** compared the mean cortical:cerebellar SUVR between the groups.

From the article

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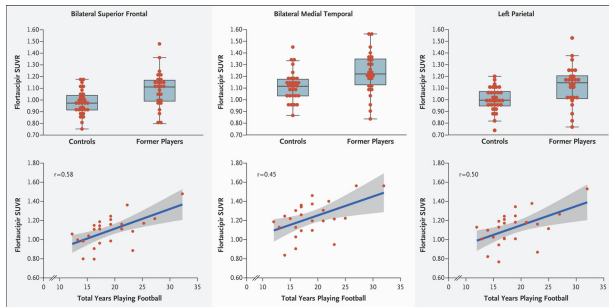
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In Part III so far:

- ▶ One sample comparison to a mean (one sample t)
- ▶ Two independent samples (two sample t)
- ▶ Two non-independent samples (paired t)
- ▶ Multiple samples/groups (ANOVA)
 - ▶ Bonferroni
 - ▶ Tukey's HSD

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But all of the methods we have looked at so far depend on some assumptions about the underlying distribution.

What have we assumed?

What do we do if our assumptions are violated?

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From <http://biostatisticsryangoslingreturns.tumblr.com/>



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PROS: Non-parametric methods make very few assumptions about the variables we sample or their distribution, and thus rely less on parameters.

- ▶ They do not use means or standard deviations
- ▶ They use a ranking of the data instead of the actual values
- ▶ They do not assume a normal distribution of the data
- ▶ They are less sensitive to outliers and skewed data
- ▶ They do not need a large sample size

CONS: Non-parametric methods use less of the information offered in the data.

- ▶ If the assumptions for a parametric test are met and a non-parametric test is used instead, the non-parametric test will have lower power (a lower probability of rejecting a false null hypothesis)
- ▶ They are less specific in what they test
- ▶ They in essence ignore important parts of the data

We will discuss non-parametric equivalents for:

Parametric test	Non-parametric equivalent
Two sample t	Wilcoxon rank-sum
Paired t	Wilcoxon signed-rank
ANOVA	Kruskal-Wallis

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Wilcoxon rank-sum

Frank Wilcoxon



In one paper in 1945 he proposed both the Wilcoxon rank-sum test and the Wilcoxon signed-rank test.

- ▶ Sometimes also called the Mann-Whitney U test
- ▶ Non-parametric test for comparing two independent samples with a continuous outcome
- ▶ This is the non-parametric counterpart of the two sample t-test
- ▶ Assumes the two distributions have the same general shape, but assumes nothing about what that shape is
- ▶ Evaluates the null hypothesis that the two populations are identical

To calculate a rank-sum test:

The observations from both groups are pooled, ordered from lowest to highest, and assigned the **rank** of their order.

If there are tied values, these are assigned the average of the ranks. For example, if two observations have the same value and the next lower value is rank 3, then the two observations are both given the rank 4.5, because they would have been ranks 4 and 5.

Then the sum of the ranks belonging to group 1 is compared to the sum of the ranks belonging to group 2.

Wilcoxon rank-sum: a small example

Values in group 1: 4, 3, 5, 2, 6

Values in group 2: 6, 5, 7, 4, 8

Pool all ten values, rank them from smallest to largest, then add up the ranks within each group.

Wilcoxon rank-sum: a small example

Group 1	rank	Group 2	rank
2	1	4	3.5
3	2	5	5.5
4	3.5	6	7.5
5	5.5	7	9
6	7.5	8	10
sum	19.5	sum	35.5

Wilcoxon rank-sum

The smaller of the two sums is called W . It is then used in the following equation to generate a Z statistic.

$$Z_w = \frac{W - \mu_w}{\sigma_w}$$

where

$$\mu_w = \frac{n_s(n_s + n_l + 1)}{2} \quad \sigma_w = \sqrt{\frac{n_s n_l (n_s + n_l + 1)}{12}}$$

Here n_s is the size of the group with the smaller rank sum and n_l is the size of the other group.

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So from our example, where group 1 had a rank sum of 19.5 and group 2 had a rank sum of 35.5:

$$\mu_w = \frac{n_s(n_s + n_l + 1)}{2} = \frac{5(5 + 5 + 1)}{2} = 27.5$$

$$\sigma_w = \sqrt{\frac{n_s n_l (n_s + n_l + 1)}{12}} = \sqrt{\frac{5 \times 5 (5 + 5 + 1)}{12}} = 4.8$$

$$Z_w = \frac{W - \mu_w}{\sigma_w} = \frac{19.5 - 27.5}{4.8} = -1.67$$

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The Z_w we generate follows an approximate standard normal distribution, so we can use our Z score to get a p-value in R.

```
2 * pnorm(-1.67)
```

```
## [1] 0.09491936
```

With a p-value of 0.09 we would not reject the null hypothesis at $\alpha = 0.05$.

The general syntax will be:

```
wilcox.test(group1, group2)
```

or

```
wilcox.test(outcome ~ group, data = mydata)
```

Unpaired is the default, so you do not need `paired = FALSE`. In current versions of R the formula version will error if you pass `paired` to it.

Remember that you can always type `help(wilcox.test)` in your console to get the full details.

Rank-sum example: phenylketonuria

Normalized mental age (nMA) scores for children with phenylketonuria, from Wrona, R.M., "A Clinical Epidemiologic Study of Hyperphenylalaninemia," *American Journal of Public Health*, volume 69, July 1979, 673-679.

Group 1: "low exposure", < 10.0 mg/dl

Group 2: "high exposure", ≥ 10.0 mg/dl

Rank-sum example: phenylketonuria

```
nMA <- c(34.5, 37.5, 39.5, 40, 45.5, 47, 47, 47.5, 48.7,  
         49, 51, 51, 52, 53, 54, 54, 55, 56.5, 57, 58.5,  
         58.5, 28, 35, 37, 37, 43.5, 44, 45.5, 46, 48,  
         48.3, 48.7, 51, 52, 53, 53, 54, 54, 55)  
Group <- c(rep("low", 21), rep("high", 18))  
pku <- data.frame(Group, nMA)  
head(pku, 3)
```

```
##   Group  nMA  
## 1   low 34.5  
## 2   low 37.5  
## 3   low 39.5
```

Rank-sum example: phenylketonuria

In this example there are 18 high exposure and 21 low exposure individuals.

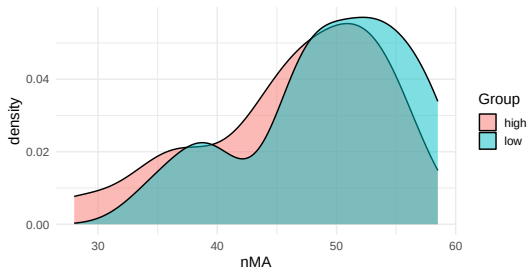
```
group_by(pku, Group) %>%  
  summarise(count = n(),  
             median = median(nMA),  
             IQR = IQR(nMA))
```

```
## # A tibble: 2 x 4  
##   Group count median   IQR  
##   <chr> <int>   <dbl> <dbl>  
## 1 high     18    48.2   9.12  
## 2 low      21    51      7
```

Rank-sum example: phenylketonuria

If we graph the distributions with a density plot, what do we notice?

```
ggplot(pku, aes(x = nMA)) +  
  geom_density(aes(fill = Group), alpha = 0.5) +  
  theme_minimal(base_size = 14)
```



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Rank-sum example: phenylketonuria

```
wilcox.test(nMA ~ Group, data = pku)

##
##  Wilcoxon rank sum test with continuity correction
##
## data:  nMA by Group
## W = 142, p-value = 0.1896
## alternative hypothesis: true location shift is not equal to 0
```

Rank-sum vs t: NHANES example

Here I will again use the NHANES data as an example, looking at height by gender.

```
nhanes <- read.csv("nhanes.csv")  
dim(nhanes)
```

```
## [1] 2503 40
```

```
head(nhanes[, c("gender", "bmxht")], 3)
```

```
##   gender bmxht  
## 1   Male 175.3  
## 2 Female 162.4  
## 3 Female 161.8
```

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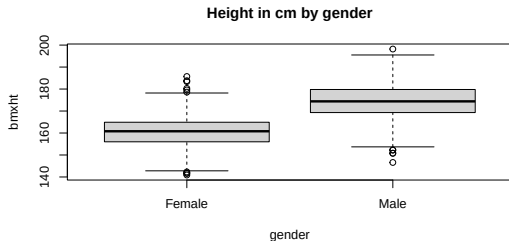
Kruskal-Wallis

Summary

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Rank-sum vs t: NHANES example

```
ht_m <- nhanes$bmght[nhanes$gender == "Male"]  
ht_f <- nhanes$bmght[nhanes$gender == "Female"]  
boxplot(bmght ~ gender, data = nhanes,  
        main = "Height in cm by gender")
```



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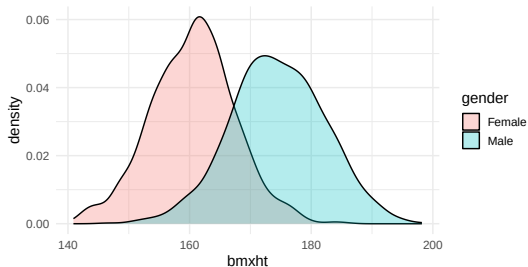
Kruskal-Wallis

Summary

Check your understanding

Rank-sum vs t

```
ggplot(nhanes, aes(x = bmxht)) +  
  geom_density(aes(fill = gender), alpha = 0.3) +  
  theme_minimal(base_size = 14)
```



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Rank-sum vs t: the parametric test

```
t.test(ht_m, ht_f, paired = FALSE)
```

```
##  
##  Welch Two Sample t-test  
##  
## data:  ht_m and ht_f  
## t = 47.285, df = 2384, p-value < 2.2e-16  
## alternative hypothesis: true difference in means is not equal to 0  
## 95 percent confidence interval:  
##  13.37441 14.53172  
## sample estimates:  
## mean of x mean of y  
## 174.4717 160.5186
```

Rank-sum vs t: the non-parametric test

```
wilcox.test(ht_m, ht_f, paired = FALSE)
```

```
##  
##  Wilcoxon rank sum test with continuity correction  
##  
## data:  ht_m and ht_f  
## W = 1402065, p-value < 2.2e-16  
## alternative hypothesis: true location shift is not equal to 0
```

When the sample size is quite large, as with these NHANES data, the assumption of approximate normality is a reasonable one, and the results of the hypothesis test will generally not differ much between the parametric and the non-parametric approach.

Both tests here give a p-value below 2.2×10^{-16} , so both reject the null.

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Wilcoxon signed-rank

- ▶ Non-parametric test for comparing two non-independent (paired) samples
- ▶ This is the non-parametric counterpart of the paired t-test
- ▶ Assumes the distribution of the differences is symmetric, but assumes nothing about what that shape is
- ▶ Evaluates the null hypothesis that the difference between the first and second measure is 0

Steps:

- 1) Calculate the difference between each pair of observations
- 2) Rank the differences by absolute value from smallest to largest. Again, tied values get the average of the ranks. Any pair where the difference is 0 is thrown out.
- 3) Assign a sign for whether the difference was positive or negative
- 4) Take the sum of the positive ranks and the sum of the negative ranks. The smaller sum is denoted T .

Under the null hypothesis that the difference is 0, we would expect the sample to have equal numbers of positive and negative ranks with equivalent sums. This expectation is tested against the statistic

$$Z_T = \frac{T - \mu_T}{\sigma_T}$$

where

$$\mu_T = \frac{n(n+1)}{4} \quad \sigma_T = \sqrt{\frac{n(n+1)(2n+1)}{24}}$$

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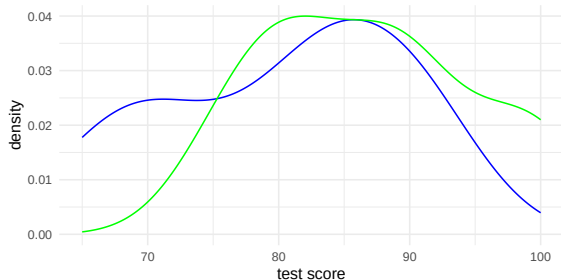
Check your understanding

Signed-rank example: pre and post test

Thirteen students each take a test twice, before and after a review session.

Time 1	Time 2	Time 1	Time 2
65	77	92	91
87	100	83	96
77	75	85	84
90	89	91	89
70	80	68	88
84	81	72	100
		81	81

Signed-rank example



Blue is time 1 and green is time 2. The scores shift up, but with only 13 students we would not want to lean on a normality assumption.

Signed-rank: calculate the difference and the sign

Time 1	Time 2	Difference	sign	Time 1	Time 2	Difference	sign
65	77	12	+	92	91	-1	-
87	100	13	+	83	96	13	+
77	75	-2	-	85	84	-1	-
90	89	-1	-	91	89	-2	-
70	80	10	+	68	88	20	+
84	81	-3	-	72	100	28	+
				81	81	0	?

Signed-rank: sort by absolute value and assign rank

Time 1	Time 2	Difference	sign	rank
90	89	-1	-	2
92	91	-1	-	2
85	84	-1	-	2
77	75	-2	-	4.5
91	89	-2	-	4.5
84	81	-3	-	6
70	80	10	+	7
65	77	12	+	8
87	100	13	+	9.5
83	96	13	+	9.5
68	88	20	+	11
72	100	28	+	12
81	81	0	?	drop

Signed-rank: sum the negative ranks

Time 1	Time 2	Difference	sign	rank
90	89	-1	-	2
92	91	-1	-	2
85	84	-1	-	2
77	75	-2	-	4.5
91	89	-2	-	4.5
84	81	-3	-	6

Sum of the negative signed ranks is 21.

Signed-rank: sum the positive ranks

Time 1	Time 2	Difference	sign	rank
70	80	10	+	7
65	77	12	+	8
87	100	13	+	9.5
83	96	13	+	9.5
68	88	20	+	11
72	100	28	+	12

Sum of the positive signed ranks is 57.

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Wilcoxon signed-rank: the example

Our expectation would be

$$\mu_T = \frac{n(n+1)}{4} = \frac{12(12+1)}{4} = 39$$

Remember that we had 13 observations, but we dropped one because the scores at times 1 and 2 were the same, so $n = 12$.

$$\sigma_T = \sqrt{\frac{n(n+1)(2n+1)}{24}} = \sqrt{\frac{12(12+1)(2 \times 12 + 1)}{24}} = 12.75$$

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Wilcoxon signed-rank: the example

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We compare our expectation to the smaller rank sum. The sum of the negative ranks was 21 and the sum of the positive ranks was 57, so $T = 21$.

$$Z_T = \frac{T - \mu_T}{\sigma_T} = \frac{21 - 39}{12.75} = -1.412$$

```
2 * pnorm(-1.412)
```

```
## [1] 0.15795
```

We do not reject the null at $\alpha = 0.05$.

Wilcoxon signed-rank in R

The general syntax will be:

```
wilcox.test(before, after, paired = TRUE)
```

or, if the two measures are columns of one data frame,

```
wilcox.test(Pair(before, after) ~ 1, data = mydata)
```

The `Pair()` form is the formula version. Note that it is `paired = TRUE` that turns the rank-sum test into the signed-rank test.

Wilcoxon signed-rank: the example in R

```
wilcox.test(test1, test2, paired = TRUE, correct = FALSE)
```

```
##  
##  Wilcoxon signed rank test  
##  
## data:  test1 and test2  
## V = 21, p-value = 0.157  
## alternative hypothesis: true location shift is not equal to 0
```

R reports 0.157 and we calculated 0.158 by hand. The tiny gap is just the rounding in our Z of -1.412.

Signed-rank: compare to the paired t

```
t.test(test1, test2, paired = TRUE)
```

```
##  
## Paired t-test  
##  
## data: test1 and test2  
## t = -2.3684, df = 12, p-value = 0.0355  
## alternative hypothesis: true mean difference is not equal to 0  
## 95 percent confidence interval:  
## -12.7011701 -0.5295991  
## sample estimates:  
## mean difference  
## -6.615385
```

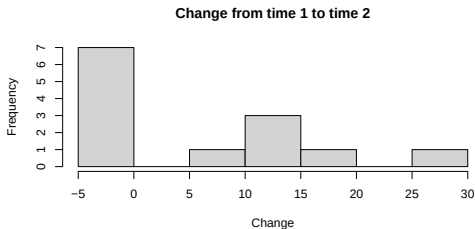
The paired t rejects at $\alpha = 0.05$ but the signed-rank test does not.

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Signed-rank: compare to the paired t

Our sample is only 13, and the distribution of the changes looks like this. Remember that the 0 difference gets thrown out of the signed-rank test.

```
hist(Change, main = "Change from time 1 to time 2")
```



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Kruskal-Wallis

The Kruskal-Wallis test is a non-parametric alternative to ANOVA.

It looks at the medians of the groups, not the means, and tests whether at least one is different from the others, but not which one.

- ▶ H_0 : there is no difference between the group medians
- ▶ H_1 : there is a statistically significant difference between the group medians

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This test can be thought of as an extension of the rank-sum test, because it is built on the same idea of ranking the pooled data.

We will not do this one by hand.

In R the syntax is generally:

```
kruskal.test(outcome ~ group, data = mydata)
```

Kruskal-Wallis

```
treatment <- c(rep("Group1", 4), rep("Group2", 5),  
               rep("Group3", 6), rep("Group4", 7))  
outcome <- c(30, 46, 99, 95,           # group 1  
             3, 14, 16, 41, 47,       # group 2  
             5, 4, 4, 4, 10, 9,      # group 3  
             24, 30, 43, 51, 62, 32, 96) # group 4  
KW_data <- data.frame(treatment, outcome)  
kruskal.test(outcome ~ treatment, data = KW_data)
```

```
##  
##  Kruskal-Wallis rank sum test  
##  
## data:  outcome by treatment  
## Kruskal-Wallis chi-squared = 13.096, df = 3, p-value =  
## 0.004434
```

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Non-parametric summary

Most parametric tests have an analogous non-parametric test. We have covered the following.

Samples	Parametric	Non-parametric
Two independent samples	two sample t-test	Wilcoxon rank-sum
Two paired samples	paired t-test	Wilcoxon signed-rank
Three or more samples	ANOVA	Kruskal-Wallis

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Samples	Test name	R function
Two independent samples	Wilcoxon rank-sum	<code>wilcox.test(g1, g2)</code>
Two paired samples	Wilcoxon signed-rank	<code>wilcox.test(g1, g2, paired = TRUE)</code>
Three or more samples	Kruskal-Wallis	<code>kruskal.test(outcome ~ group)</code>

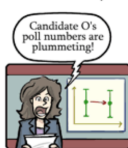
The formula version of `wilcox.test()` does not take a `paired` argument. Use the two-vector form for paired data, or `wilcox.test(Pair(before, after) ~ 1)`.

Parting humor (from phdcomics.com)

Dear News Media,

When reporting poll results, please keep in mind the following suggestions:

1. If two poll numbers differ by less than the margin of error, it's not a news story.



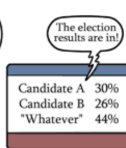
2. Scientific facts are not determined by public opinion polls.



3. A poll taken of your viewers/internet users is not a scientific poll.



4. What if all polls included the option "Don't care"?



Signed,

-Someone who took a
basic statistics course.

JORGE CHAM © 2010

WWW.PHDCOMICS.COM

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In which of these scenarios would we want to use a non-parametric test?

1. Jessica collects data on student study habits, but her sample is too small to lean on the central limit theorem.
2. Chandler compares midterm scores for students who did and did not take a practice midterm. Both distributions are approximately normal.
3. Mitchell and Antonia relate sleep hours to PH142 performance. Sleep hours are left skewed with outliers on both sides, and they have only 20 students.
4. Alma and Yunzhe find the distribution of homework scores is bimodal.

Check your understanding: answer

Scenarios 1, 3, and 4.

1. **Yes.** With a small sample we cannot appeal to the CLT for normality of the sampling distribution, and rank-based tests do not need it.
2. **No.** Both groups are approximately normal, so the parametric two sample t-test is appropriate and has more power.
3. **Yes.** Skew plus outliers plus $n = 20$ is exactly the case rank-based tests were built for.
4. **Yes.** A bimodal distribution is nowhere near normal, and the mean is a poor summary of it.

Check your understanding: COVID antibody data

A study by Jalkanen et al. examined the effectiveness of the BNT162b2 mRNA COVID-19 vaccine.

One indicator of effectiveness is the concentration of the anti-S1 IgG antibody. One group of volunteers was measured before vaccination and a different group of volunteers was measured after vaccination.

The researchers want to know whether there is a difference in IgG antibodies between the two samples.

Which test should we use, and what are the missing ranks?

Check your understanding: the data

0 days	Rank	42 days	Rank
30	a	111	b
26	6	112	12
23	4	107	8
21	3	111	d
19	c	127	14
17	1	108	9
25	5	115	13
Sum	e	Sum	f

Check your understanding: answer

The two groups are **different volunteers**, so the samples are independent and we use the **Wilcoxon rank-sum test**.

Pool all 14 values and rank them from smallest to largest.

$a = 7$, $b = 10.5$, $c = 2$, $d = 10.5$

The two 111 values tie for ranks 10 and 11, so both get 10.5.

$e = 28$, $f = 77$

As a check, the ranks must sum to $\frac{14 \times 15}{2} = 105$, and $28 + 77 = 105$.

Check your understanding: answer

$W = 28$ is the smaller sum, with $n_s = n_l = 7$.

$$\mu_w = \frac{7(7 + 7 + 1)}{2} = 52.5 \quad \sigma_w = \sqrt{\frac{7 \times 7 \times 15}{12}} = 7.83$$

$$Z_w = \frac{28 - 52.5}{7.83} = -3.13$$

```
2 * pnorm(-3.13)
```

```
## [1] 0.001748063
```

We reject the null. IgG concentrations differ before and after vaccination.